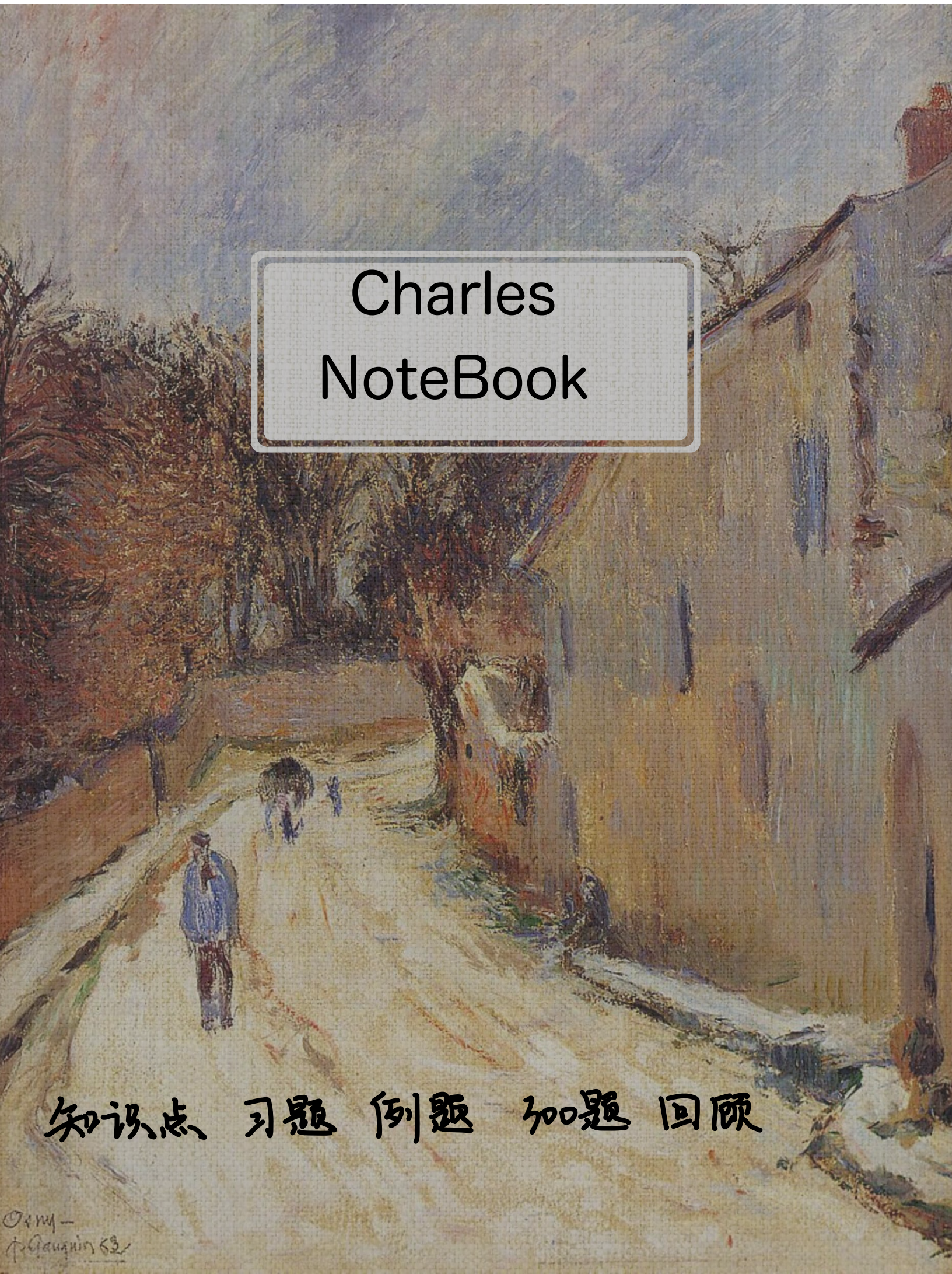


An impressionist painting of a street scene. In the foreground, a man in a blue jacket and dark pants stands on a light-colored path. To his right, a large, leafy tree with brown and orange foliage stands. In the background, more trees and a building with a red roof are visible. The sky is a mix of blue and white. The overall style is loose and painterly, with visible brushstrokes.

Charles NoteBook

知识点 习题 例题 加题 回顾

第七讲 零点问题和微分不等式

- ① { 零点问题
方程的根
曲线交点
- ② { 微分不等式

题型一 零点根

1. 零点定理 (证明根存在性)

$$\left\{ \begin{array}{l} [a, b] \text{ 连续} \\ f(a)f(b) < 0 \\ \lim_{x \rightarrow a^+} f(x) = \alpha, \lim_{x \rightarrow b^-} f(x) = \beta, \alpha\beta < 0 \end{array} \right\} \Rightarrow \text{至少一根}$$

a. 单调性 (证明根的唯一性)

$$\left\{ \begin{array}{l} f(x) \text{ 在 } (a, b) \text{ 单调} \\ f'(x) > 0 \text{ 或 } f'(x) \text{ 存在且 } f'(x) \neq 0 \end{array} \right\} \Rightarrow \text{至多一根}$$

例 7.4

设常数 $k > 0$, $f(x) = \ln x - \frac{x}{e} + k$ 在 $(0, +\infty)$ 零点个数

$$f'(x) = \frac{1}{x} - \frac{1}{e} \stackrel{?}{=} 0 \Rightarrow x = e$$

x	$(0, e)$	e	$(e, +\infty)$
$f'(x)$	+	0	-
$f(x)$	$\nearrow$	$k > 0$	$\searrow$

2

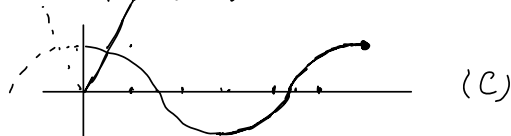
例 7.5

在 $(-\infty, +\infty)$ 内, 方程 $|x|^{\frac{1}{4}} + |x|^{\frac{1}{2}} - \cos x = 0$

(A) 无实根 (B) 有且仅有一个实根 (C) 有且仅有两个实根 (D) 有无穷多个实根

$$f(x) = |x|^{\frac{1}{4}} + |x|^{\frac{1}{2}} - \cos x$$

$$\text{设 } x \geq 0, f(x) = x^{\frac{1}{4}} + x^{\frac{1}{2}} - \cos x, f'(x) = \frac{1}{4}x^{-\frac{3}{4}} + \frac{1}{2}x^{-\frac{1}{2}} + \sin x$$



例 7.6

讨论 $y = x \ln x$ 与 $y = -A$ 交点个数

$$\lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} \frac{\ln x}{\frac{1}{x}} = \lim_{x \rightarrow 0} \frac{\frac{1}{x}}{-\frac{1}{x^2}} = 0$$

$$y = \ln x + 1 \stackrel{?}{=} 0 \Rightarrow x_0 = \frac{1}{e}$$

$$A > \frac{1}{e} \Rightarrow 0 \text{ 个交点} \quad A = \frac{1}{e} \Rightarrow 1 \text{ 个交点} \quad 0 < A < \frac{1}{e} \Rightarrow 2 \text{ 个交点}$$

x	0	$(0, \frac{1}{e})$	$\frac{1}{e}$	$(\frac{1}{e}, +\infty)$
$f'(x)$	-	0	+	
$f(x)$	$\searrow$	$\nearrow$	$-\frac{1}{e}$	$\nearrow$

例 7.7

求 $k \arctan x - x = 0$ 的不同实根的个数, k 为参数

$$f(x) = k \arctan x - x$$

$$f'(x) = \frac{k}{1+x^2} - 1 \stackrel{?}{=} 0 \quad k = 1+x^2 \quad x^2 = k-1$$

① 当 $k \leq 1$ 时 $f'(x) \leq 0$ $f(x) \downarrow$ ($f'(x) = 0$, 当且仅当 $x=0$)

$$\lim_{x \rightarrow -\infty} f(x) = +\infty \quad \lim_{x \rightarrow +\infty} f(x) = -\infty \quad \text{有且仅有一个根}$$

② 当 $k > 1$ 时 $f'(x) = 0$ 时: $x = \pm \sqrt{k-1}$

$f(x)$ 是奇函数, 仅讨论 $x > 0$ 时

x	0	$(0, \sqrt{k-1})$	$\sqrt{k-1}$	$(\sqrt{k-1}, +\infty)$	$\lim_{x \rightarrow +\infty} \frac{k}{1+x^2} - 1 < 0$
$f'(x)$		-	0	+	
$f(x)$	0	$\downarrow$		$\uparrow$	

$\therefore$ 一共 3 个零点.

题型二

罗尔定理

3. 罗尔定理 (只证不证)

若 $f^{(n)}(x) = 0$ 至多有 k 个根, 则 $f(x) = 0$ 至多有 $k+n$ 个根

$\Rightarrow f^{(n)}(x) \neq 0 \Rightarrow f^{(n-1)}(x) = 0$ 至多 n 个根 $\Rightarrow f(x) = 0$ 至多 n 个根

题型三

实系数奇次方

程至少一实根

4. 实系数奇次方程至少一实根

证: $x^{2n+1} + a_1 x^{2n} + \dots + a_{2n} x + a_{2n+1} = 0$ 至少有一个实根

$$\text{设 } f(x) = x^{2n+1} + a_1 x^{2n} + \dots + a_{2n} x + a_{2n+1}$$

$$x \rightarrow -\infty \quad x \rightarrow +\infty$$

$$f(x) \rightarrow -\infty \quad f(x) \rightarrow +\infty \Rightarrow f(x) = 0$$

例 7.1

证明 $x + p + q \cos x = 0$ 恰有一个实根 $0 < q < 1$

$$f(x) = x + p + q \cos x$$

$$f'(x) = 1 - q \sin x > 0$$

$$\lim_{x \rightarrow -\infty} f(x) = -\infty \quad \lim_{x \rightarrow +\infty} f(x) = +\infty \quad \text{由广义零点定理} \Rightarrow f(x) = 0$$

例 7.2

若 $f(x) = (x-1)(x-2)(x-3)(x-4)$, 则 $f(x)$ 零点个数

$f(x)$ 零点个数为 4 $\Rightarrow f'(x) \dots \dots \dots$ 3 (至少)
 $(x^4 f^{(4)}) \neq 0 \Rightarrow f'(x) \dots \dots \dots$ 3 (至多)

例 7.3

若 $3a^2 - 5b < 0$, 则方程 $x^5 + 2ax^3 + 3bx + 4c = 0$

① $f(x) = x^5 + 2ax^3 + 3bx + 4c$

$f'(x) = 5x^4 + 6ax^2 + 3b$

$g(x) = 5x^2 + 6ax + 3b \quad \Delta = 36a^2 - 60b = 12(3a^2 - 5b) < 0$

$\Rightarrow g(x) > 0$

$f'(x)$ 至多 0 个根 $\Rightarrow f(x)$ 至多 1 个根

② $f(x)$ 是奇次多项式 $\Rightarrow f(x)$ 至少 1 个根 | 1 个根

= 微分不等式

题型四

用函数性质

例 7.8

证明 当 $x > 0$ 时, $\ln(1+\frac{1}{x}) < \frac{1}{\sqrt{x(x+1)}}$

~~$f(x) = \ln(1+\frac{1}{x}) - \frac{1}{\sqrt{x(x+1)}}$~~

~~$= \ln(x+1) - \ln x - [x(x+1)]^{-\frac{1}{2}}$~~

~~$f'(x) = \frac{1}{x+1} - \frac{1}{x} + \frac{1}{2}[x(x+1)]^{-\frac{3}{2}} \cdot (2x+1)$~~

~~$= \frac{1}{x(x+1)} + \frac{2x+1}{2[x(x+1)]^{\frac{3}{2}}} > 0$~~

化简!

$\Rightarrow$ 证 $\sqrt{x(x+1)} \ln(1+\frac{1}{x}) < 1$

$\xrightarrow{x=t} \sqrt{1+t} \ln(1+t) - t < 0$

令 $f(t) = \sqrt{1+t} \ln(1+t) - t$

$f'(t) = \frac{1}{2}(1+t)^{-\frac{1}{2}} \ln(1+t) + \frac{\sqrt{1+t}}{1+t} - 1$

$= \frac{\ln(1+t) + 2 - 2\sqrt{1+t}}{2\sqrt{1+t}}$

令 $g(t) = \ln(1+t) + 2 - 2\sqrt{1+t}$

$g'(t) = \frac{1}{1+t} - \frac{1}{\sqrt{1+t}} < 0$

$\therefore f'(t) < 0 \Rightarrow f(t) \downarrow \Rightarrow f(t) < f(0) = 0$

例 7.9 (用凹凸性)

证明 当 $0 < x < \frac{\pi}{2}$ 时, 有 $\sin x > \frac{2x}{\pi}$

$$\text{设 } f(x) = \sin x - \frac{2x}{\pi}$$

$$f'(x) = \cos x - \frac{2}{\pi} \stackrel{?}{\leq} 0 \quad \cos x \leq \frac{2}{\pi}$$

$$f''(x) = -\sin x < 0 \quad \square$$

$$\begin{array}{ccc} x & 0 & (\frac{\pi}{2}, \frac{\pi}{2}) \\ f'(x) & 0 & - \\ f(x) & 0 & \nearrow \end{array}$$

例 7.10 (用极值)

证明 $1 + x \ln(x + \sqrt{1+x^2}) \geq \sqrt{1+x^2}$, $-\infty < x < +\infty$

$$\text{令 } f(x) = 1 + x \ln(x + \sqrt{1+x^2}) - \sqrt{1+x^2}$$

$$f'(x) = \ln(x + \sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} - \frac{x}{\sqrt{1+x^2}} = \ln(x + \sqrt{1+x^2}) \stackrel{?}{\geq} 0$$

$$x_0 + \sqrt{1+x_0^2} = 1 \Rightarrow x_0 = 0.$$

$$f''(x) = \frac{1}{1+x^2} > 0 \quad \square$$

唯一极小值 即为最小值 $f(0) = 0$

题型五

用参数变量化

例 7.11

设 $0 < a < b$, 证明 $\ln \frac{b}{a} > 2 \frac{b-a}{a+b}$

$$\Rightarrow \text{可以证: } \ln \frac{x}{a} > 2 \frac{x-a}{x+a}$$

$$\text{可以证: } \ln \frac{b}{x} > 2 \frac{b-x}{x+b}$$

$$\text{可以证: } \ln \frac{b}{a} > 2 \frac{\frac{b}{a}-1}{\frac{b}{a}+1}, \quad x = \frac{b}{a}, \quad x > 1$$

$$\text{证: } \ln x > 2 \frac{x-1}{x+1}$$

$$f(x) = f'(x) = f''(x)$$

题型六

用中值定理或泰勒

$$\left\{ \begin{array}{l} f(b) - f(a) = f'(\xi)(b-a) \\ f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\eta)}{2!}(x-x_0)^2 \end{array} \right.$$

例 7.12

设 $0 < a < b < 1$ 证明 $\arctan b - \arctan a < \frac{b-a}{2ab}$

$$\text{设 } f(x) = \arctan x$$

$$f(b) - f(a) = \frac{b-a}{1+\xi^2} \quad \xi \in (a, b)$$

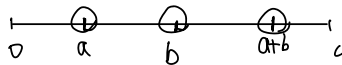
$$a < \xi < b \Rightarrow \frac{1}{1+b^2} < \frac{1}{1+\xi^2} < \frac{1}{1+a^2}$$

$$\Rightarrow \arctan b - \arctan a < \frac{b-a}{1+q^2} < \frac{b-a}{b^2+q^2} < \frac{b-a}{2ab}$$

例 7.13

设 $f(x)$ 在闭区间 $[0, c]$ 上连续, 其导数 $f'(x)$ 在开区间 $(0, c)$ 内存在且单调减少, 又 $f(0)=0$, 用拉氏证明 $f(a+b) \leq f(a) + f(b)$, 其中 $a, b: 0 \leq a \leq b \leq a+b \leq c$

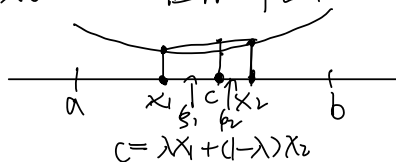
证明:



$$\begin{aligned} ① \quad & f(a) - f(0) = f'(\xi_1)(a-0) \\ & \Rightarrow f(a) = f'(\xi_1) \cdot a \\ ② \quad & f(a+b) - f(b) = f'(\xi_2)(a+b-b) \\ & \Rightarrow f(a+b) - f(b) = f'(\xi_2) \cdot a \\ ③ \quad & f'(x) < 0 \Rightarrow f'(\xi_1) > f'(\xi_2) \Rightarrow a f'(\xi_1) > a f'(\xi_2) \\ & \Rightarrow f(a) > f(a+b) - f(b) \end{aligned}$$

例 7.14

设 $f(x)$ 在 (a, b) 内二阶可导, 且 $f''(x) > 0$, 证明 $\forall x_1, x_2 \in (a, b) \quad x_1 \neq x_2$ 且 $\lambda \in (0, 1)$ 恒有 $f[\lambda x_1 + (1-\lambda)x_2] < \lambda f(x_1) + (1-\lambda)f(x_2)$



$$\begin{cases} f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f''(\xi)}{2}(x-x_0)^2 \\ f(x_1) = f(c) + f'(c)(x_1-c) + \frac{f''(\xi_1)}{2}(x_1-c)^2 \\ f(x_2) = f(c) + f'(c)(x_2-c) + \frac{f''(\xi_2)}{2}(x_2-c)^2 \end{cases}$$

$$\Rightarrow \begin{cases} \lambda f(x_1) = \lambda f(c) + \lambda f'(c)(x_1-c) + \lambda \frac{f''(\xi_1)}{2}(x_1-c)^2 \\ (1-\lambda)f(x_2) = (1-\lambda)f(c) + (1-\lambda)f'(c)(x_2-c) + (1-\lambda)\frac{f''(\xi_2)}{2}(x_2-c)^2 \end{cases}$$

$$\begin{aligned} & \lambda f'(c)(x_1-c) + (1-\lambda)f'(c)(x_2-c) \\ &= f'(c) [\lambda x_1 - \lambda c + x_2 - c - \lambda x_2 + \lambda c] \\ &= f'(c) [\lambda x_1 + (1-\lambda)x_2 - c] = 0 \end{aligned}$$

$$\Rightarrow \lambda f(x_1) + (1-\lambda)f(x_2) = f(c) + \lambda \frac{f''(\xi_1)}{2}(x_1-c)^2 + (1-\lambda)\frac{f''(\xi_2)}{2}(x_2-c)^2 > f(c)$$

习 7.1

设 $x > 0$ 时, 方程 $kx + \frac{1}{x^2} = 1$ 有且仅有一个根. 求 k 取值范围

$$f(x) = kx + \frac{1}{x^2} - 1$$

$$f'(x) = k - 2x^{-3}$$

$$f''(x) = 6x^{-4} > 0$$

① $f'(x) \equiv 0 \quad x^3 k = 2$

$$\Rightarrow x = \sqrt[3]{\frac{2}{k}} \quad (k \neq 0) > 0 \Rightarrow k > 0.$$

$$f'(\sqrt[3]{\frac{2}{k}}) = 0$$



$$\Rightarrow (\frac{2}{k})^{\frac{1}{3}} k + (\frac{2}{k})^{\frac{2}{3}} - 1 = 0$$

$$2 k^{\frac{2}{3}} + k^{\frac{2}{3}} = 2^{\frac{1}{3}}$$

$$k^{\frac{2}{3}} = \frac{2^{\frac{1}{3}}}{3} \quad k = \frac{2}{(\frac{2}{3})^3} = \frac{2}{9} \sqrt{3}$$

② $k \leq 0$ 时.

$$\lim_{x \rightarrow 0} f(x)$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (kx + \frac{1}{x^2} - 1) = +\infty$$

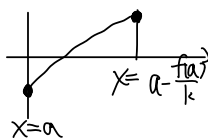
$$\lim_{x \rightarrow \infty} f(x)$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} (kx + \frac{1}{x^2} - 1) = \begin{cases} -\infty & k \neq 0 \\ -1 & k = 0 \end{cases}$$

$\therefore k \leq 0$ 及 $k = \frac{2}{9} \sqrt{3}$ 时 在 $(0, +\infty)$ 仅一根

习 7.2

设 $f(x)$ 在 $[a, +\infty)$ 上连续, 且当 $x > a$ 时, $f'(x) > k > 0$, 其中 k 为常数. 若 $f(a) < 0$, 证明方程 $f(x) = 0$ 在 $[a, a - \frac{f(a)}{k}]$ 内有且仅有一个实根



证明 $f(a - \frac{f(a)}{k}) = 0$

$$f(a - \frac{f(a)}{k}) - f(a) = (-\frac{f(a)}{k}) \cdot (f'(\xi)) > -f(a)$$

$$\Rightarrow f(a - \frac{f(a)}{k}) > 0 \quad \text{零点定理} \Rightarrow \text{得证}$$

② $f'(x) > 0 \Rightarrow$ 单调

习 7.3

证明对任意实数 a, b , 且 $a < b$ 都有 $\sin b - \sin a \leq b - a$

$$f(x) = \sin x$$

$$f(b) - f(a) = (b - a) f'(\xi)$$

$$\text{证: } \frac{\sin b - \sin a}{b - a} \leq 1$$

$$\frac{\sin b - \sin a}{b - a} = \frac{f'(\xi)}{\cos \xi} \leq 1 \quad \text{得证}$$

习7.4

证明对任意 $x \in (-\infty, +\infty)$, 都有 $x - x^2 < \frac{1}{e}$

$$f(x) = x - x^2 - \frac{1}{e}$$

$$f'(x) = 1 - 2x \stackrel{=} {=} 0 \Rightarrow x = \frac{1}{2}$$

$$x \quad (-\infty, \frac{1}{2}) \quad \frac{1}{2} \quad (\frac{1}{2}, +\infty)$$

$$\begin{array}{ccc} f(x) & + & 0 & - \\ f'(x) & \nearrow & \frac{1}{e} & \searrow \end{array}$$

$$f(\frac{1}{2}) < 0 \quad \text{得证}$$

习7.5

设 $b > a > e$, 证明 $a^b > b^a$

$$e^{b \ln a} > e^{a \ln b}$$

$$b \ln a > a \ln b$$

$$\frac{b}{a} > \ln \frac{b}{a}$$

$$b > a > e \Rightarrow \frac{b}{a} > 1 \Rightarrow f(x) = x - \ln x, \quad x \in (1, +\infty)$$

$$f'(x) = 1 - \frac{1}{x} \stackrel{=} {=} 0 \Rightarrow x = 1$$

$$f''(x) = \frac{1}{x^2} > 0$$

$$\min(f(x)) = 1 > 0$$

$$\therefore \frac{b}{a} > \ln \frac{b}{a}$$

习7.6

证明 $(\ln \frac{1+x}{x} - \frac{1}{1+x})^2 < \frac{1}{x(1+x)^2} \quad (x > 0)$

$$|\ln \frac{1+x}{x} - \frac{1}{1+x}| < \frac{1}{(1+x)\sqrt{x}} \quad (x > 0)$$

$$g(x) = \ln \frac{1+x}{x} - \frac{1}{1+x}$$

$$g'(x) = \frac{1}{1+x} \cdot \frac{1}{x^2} + \frac{1}{(1+x)^2} = \frac{1}{x(1+x)^2} > 0$$

$$\lim_{x \rightarrow 0^+} g(x) = +\infty$$

$$\lim_{x \rightarrow +\infty} g(x) = 0$$

$$g(x) > 0$$

$$\Rightarrow \text{证: } \ln \frac{1+x}{x} - \frac{1}{1+x} < \frac{1}{\sqrt{x}(1+x)}$$

$$\text{设 } f(x) = \ln \frac{1+x}{x} - \frac{1}{1+x} - \frac{1}{\sqrt{x}(1+x)}$$

$$f'(x) = \frac{-1}{x(1+x)} + \frac{1}{(1+x)^2} + \frac{\frac{1}{2}x^{-\frac{3}{2}}(1+x) + \sqrt{x}}{x(1+x)^2}$$

$$= \frac{-(1+x) + x + \frac{1}{2}x^{-\frac{1}{2}}(1+x) + \sqrt{x}}{x(1+x)^2}$$

$$-2\sqrt{x} + 1 + 3\sqrt{x}$$

$$= \frac{-\sqrt{x} - \frac{\sqrt{x}}{2} + \frac{1}{2} + \frac{x}{\sqrt{x}} + \sqrt{x}}{\sqrt{x}(1+x)^2} > 0$$

$$\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} \ln \frac{1+x}{x} - \frac{1}{\sqrt{x}(1+x)}$$

$$= \lim_{x \rightarrow +\infty} \ln(1 + \frac{1}{x}) - \frac{1}{x^{\frac{3}{2}}} = \lim_{t \rightarrow 0^+} t - t^{\frac{3}{2}} < 0$$

$$\therefore f(x) < 0$$

300.7.1

证明 $\frac{1}{x-1} + \frac{2}{x-2} + \frac{3}{x-3} = 0$ 有两实根

$$\frac{1}{x-1} + \frac{2}{x-2} + \frac{3}{x-3} = 0$$

$$(x-2)(x-3) + 2(x-1)(x-3) + 3(x-1)(x-2) = 0$$

$$f(x) = (x-2)(x-3) + 2(x-1)(x-3) + 3(x-1)(x-2)$$

$$f(1) > 0 \quad f(2) < 0 \quad f(3) > 0$$

有一个 有一个

300.7.2

$$\frac{a_n}{n+1} + \frac{a_{n-1}}{n} + \dots + a_0 = 0$$

证明方程 $a_n x^n + a_{n-1} x^{n-1} + \dots + a_0 = 0$ 在 $(0,1)$ 至少有一个实根

$$f(x) = \frac{a_n x^{n+1}}{n+1} + \frac{a_{n-1} x^n}{n} + \dots + a_0 x$$

$$f'(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$$

$$\therefore \frac{a_n}{n+1} + \frac{a_{n-1}}{n} + \dots + a_0 = 0$$

$$\therefore f(1) = 0 = f(0) \Rightarrow \text{罗尔, } f'(\xi) = 0 \quad \xi \in (0,1)$$

300.7.3

$$[1, +\infty) \text{ 上, } f''(x) < 0, \quad f(1) = 2, \quad f'(1) = -3$$

证明: $f(x) = 0$ 在 $(1, +\infty)$ 内只有一个实根

$$f''(x) < 0 \Rightarrow f(x) < -3, \quad x > 1$$

$$f'(1) = -3 \Rightarrow f(x) \text{ 在 } [1, +\infty) \text{ 上 } \downarrow \text{ 至多一个实根}$$

$$f(2) - f(1) = f'(\xi)(2-1) = f'(\xi) < -3$$

$$\Rightarrow f(2) < 2-3 = -1 \Rightarrow \text{至少一个实根}$$

$$f(1) = 2$$

300.7.4

当 a 取下列哪个值时, $f(x) = 2x^3 - 9x^2 + 12x - a$ 恰有两个不同的零点,

(A) 2 (B) 4 (C) 6 (D) 8

$$f(x) = 2x^3 - 9x^2 + 12x - a$$

$$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x-2)(x-1) \stackrel{\geq}{=} 0$$

$$x_1 = 1 \quad x_2 = 2$$

$$\textcircled{1} f(1) = 2 - 9 + 12 - a = 0 \Rightarrow a_1 = 5$$

$$\textcircled{2} f(2) = 16 - 36 + 24 - a = 0 \Rightarrow a_2 = 4$$

300.7.5

证明不等式 $\ln(1+\frac{1}{x}) > \frac{1}{1+x}$ ($0 < x < +\infty$)

$$\ln(1+\frac{1}{x}) > \frac{1}{1+x}$$

$$f(t) = \ln(1+t) - \frac{t}{1+t}, \quad t \in (0, +\infty)$$

$$f'(t) = \frac{1}{1+t} - \frac{1}{(1+t)^2} = \frac{t}{(1+t)^2} > 0$$

$$\lim_{t \rightarrow 0^+} f(t) = \lim_{t \rightarrow 0^+} \frac{(1+t)\ln(1+t) - t}{1+t}$$

$$= \lim_{t \rightarrow 0^+} \ln(1+t) + t \ln(1+t) - t = \lim_{t \rightarrow 0^+} \left[t - \frac{t^2}{2} + o(t^2) \right] + \left[t^2 - o(t^2) \right] - t = \lim_{t \rightarrow 0^+} \frac{t^2}{2} + o(t^2) > 0$$

300.7.6

证明: 当 $0 < a < b < \pi$ 时,

$$b \sin b + 2 \cos b + \pi b > a \sin a + 2 \cos a + \pi a$$

$$f(x) = x \sin x + 2 \cos x + \pi x$$

$$\text{证: } f(b) > f(a) \Rightarrow f(x) \text{ 在 } (0, \pi) \uparrow$$

$$f'(x) = x \cos x - \sin x + \pi$$

$$f''(x) = \cos x - x \sin x - \cos x = -x \sin x < 0$$

$$f'(x) \downarrow$$

$$\lim_{x \rightarrow \pi} x \sin x + 2 \cos x + \pi x > 0 \Rightarrow f'(x) > 0 \quad f(x) \uparrow$$

300.7.7

证明不等式

(1) 当 $x \in (0, \frac{\pi}{2})$ 时, $x < \tan x$

$$f(x) = x - \tan x$$

$$f'(x) = 1 - \frac{1}{1+x^2} = \frac{x^2}{1+x^2} > 0$$

$$f(\frac{\pi}{2}) = \frac{\pi}{2} - 1 > 0$$

(2) 当 $x \in (0, \frac{\pi}{2})$ 时, $(x^2 + \sqrt{4+x^4}) \cos x < 2$

$$f(x) = (x^2 + \sqrt{4+x^4}) \cos x - 2$$

$$\geq (x^2 + \sqrt{8x^4}) \cos x - 2 = (1+2\sqrt{2})x^2 \cos x - 2$$

$$f'(x) = -\sin x (x^2 + \sqrt{4+x^4}) + (2x + \frac{2x^3}{\sqrt{4+x^4}}) \cos x$$

$$= \left(2x + \frac{2x^3}{\sqrt{4+x^4}} - \tan x (x^2 + \sqrt{4+x^4}) \right) \cos x$$

$$= \left[2x + \frac{x^3}{\sqrt{4+x^4}} - \frac{\sqrt{4+x^4} \tan x (x^2 + \sqrt{4+x^4})}{x} \right] \cos x$$

$$= \left(1 + \frac{x^2}{\sqrt{4+x^4}} \right) \left(2x - \tan x \sqrt{4+x^4} \right) \cos x$$

$$= x (\sqrt{4+x^4} + x^2) \left(\frac{2}{\sqrt{4+x^4}} - \frac{\tan x}{x} \right) \cos x < 0$$

$$\frac{\tan x}{x} > 1 \quad \frac{2}{1+x^4} < 1$$

$$f(x) \downarrow$$

$$f(0) = (0+2)(-2) = 0$$